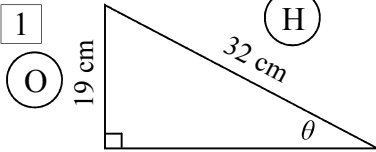


Trigonometry - Elevation/Depression - Finding an Angle (Sheet 1)

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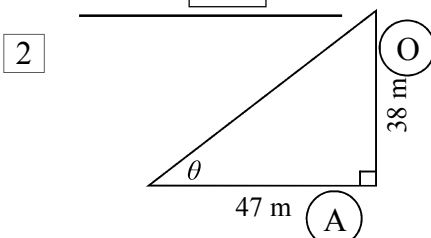
Find the angle 'θ' to the nearest degree.



$$\sin \theta = \frac{\text{O}}{\text{H}}$$

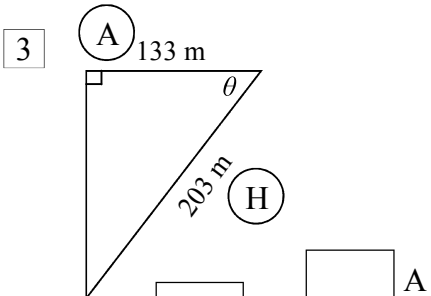
Calc: `SHIFT SIN 19 ÷ 32 =`

$$\theta = \boxed{}^\circ$$



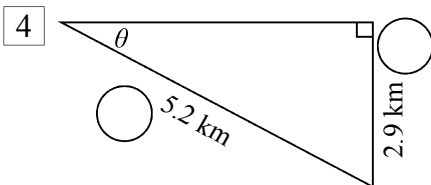
$$\theta = \frac{\text{O}}{\text{A}}$$

$$\theta = \boxed{}^\circ$$



$$\theta = \frac{\text{A}}{\text{H}}$$

$$\theta = \boxed{}^\circ$$

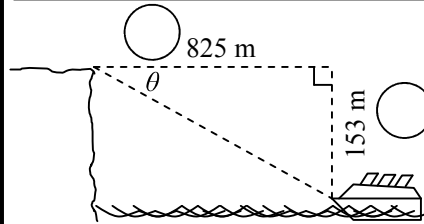


$$\theta = \frac{\text{O}}{\text{H}}$$

$$\theta = \boxed{}^\circ$$

SOH CAH TOA

A ship is 825 m from a cliff, which is 153 m tall. Eric is at the top of the cliff, looking down at the ship.

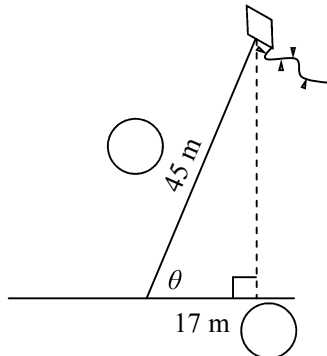


5 Calculate the angle of depression

$$\theta = \frac{\text{O}}{\text{H}}$$

$$\theta = \boxed{}^\circ$$

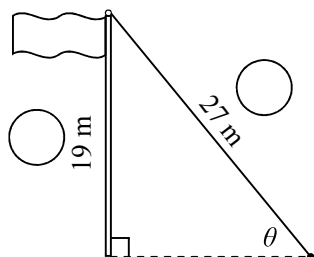
An airborne kite is attached to the ground. Its string is 45 m long and is fully extended. The kite is flying above a point 17 m away.



6 Calculate the angle of elevation from the anchor point up to the kite.

$$\theta = \frac{\text{O}}{\text{H}}$$

A flag pole has a guy wire attached to reduce vibration. The flagpole is 19 m tall and the guy wire is 27 m long.

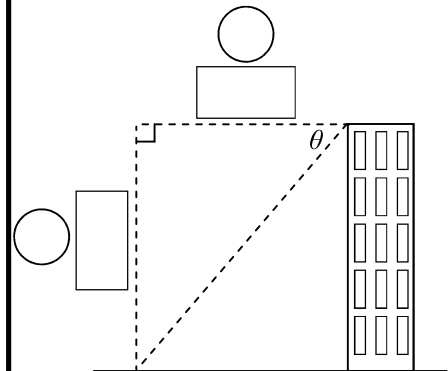


7 Calculate the angle of elevation from the anchor point up to the pole.

$$\theta = \frac{\text{O}}{\text{H}}$$

Tom is on top of a 63 m tall building, looking down at Fred, who is 48 m away from the base of the building.

8 Write the distances in the boxes.



9 Calculate the angle of depression

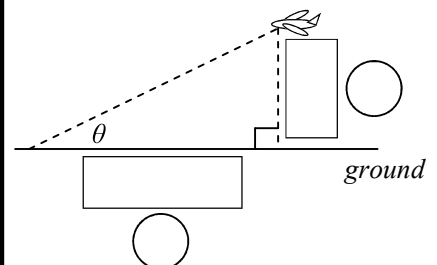
$$\theta = \frac{\text{O}}{\text{H}}$$

Yve is looking up at a plane which is 3.7 km away and is flying at a height of 800 m.

10 Convert 3.7 km to metres

$$3.7 \text{ km} = \boxed{} \text{ m}$$

11 Write the distances in the boxes.



12 Calculate the angle of elevation

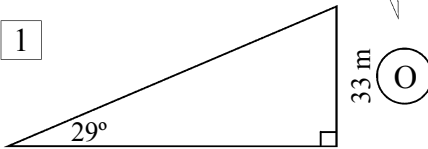
$$\theta = \frac{\text{O}}{\text{H}}$$

Trigonometry - Elevation/Depression - Finding a Length (Sheet 2)

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Find the length 'x', rounding to 1 decimal place.

1



$$\tan \square = \frac{\square}{\square}$$

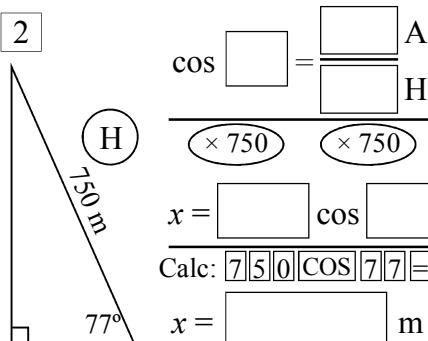
swap swap

$$x = \frac{\square}{\tan \square}$$

Calc: 33 ÷ TAN 29 =

$$x = \square \text{ m } 1 \text{ d.p.}$$

2



$$\cos \square = \frac{\square}{\square}$$

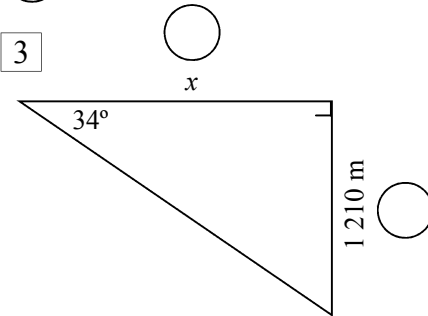
× 750 × 750

$$x = \square \cos \square$$

Calc: 750 × COS 77 =

$$x = \square \text{ m } 1 \text{ d.p.}$$

3



$$\tan \square = \frac{\square}{\square}$$

swap swap

$$x = \frac{\square}{\tan \square}$$

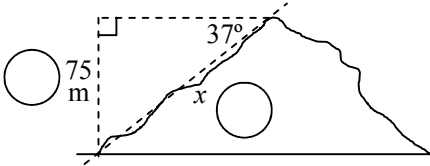
$$x = \square \text{ m } 1 \text{ d.p.}$$

SOH

CAH

TOA

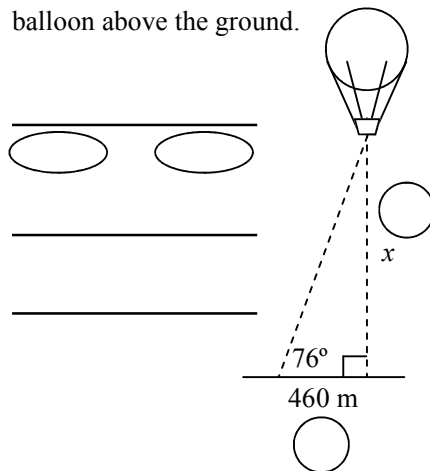
Jill is on top of a hill which has a height of 75 m. The angle of depression from the top to the base of the hill is 37°.



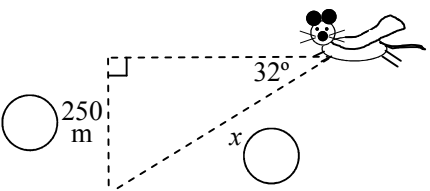
4 Estimate the distance, 'x', Jill travelled to climb the hill.

Ted is looking up at a hot air balloon. The angle of elevation is 76° and the balloon is above a point 460 m away.

5 Calculate 'x', the height of the balloon above the ground.

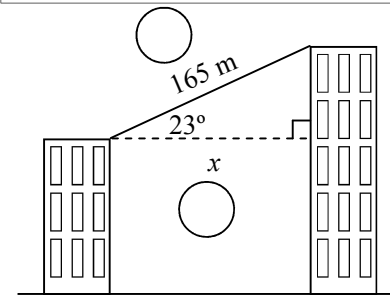


Flighty Mouse, the flying super mouse spies a damsel in distress. The angle of depression to the damsel is 32° and Flighty is flying at a height of 250 m above the ground.



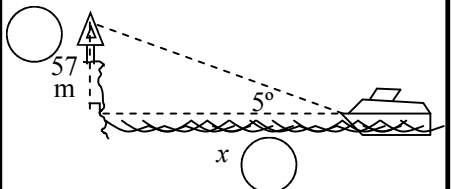
6 Calculate 'x', the flying distance Flighty Mouse has to cover.

Balancing Bec is set to walk a 165 m long tight-rope suspended between two buildings. The angle of elevation from the lower building is 23°.



7 Find 'x', the distance between the two buildings.

Stinky Stew the tug boat captain, spots a cliff beacon at an angle of elevation of 5°. He knows the beacon is 57 m above sea level.



8 Find 'x', the distance to the cliff.
